

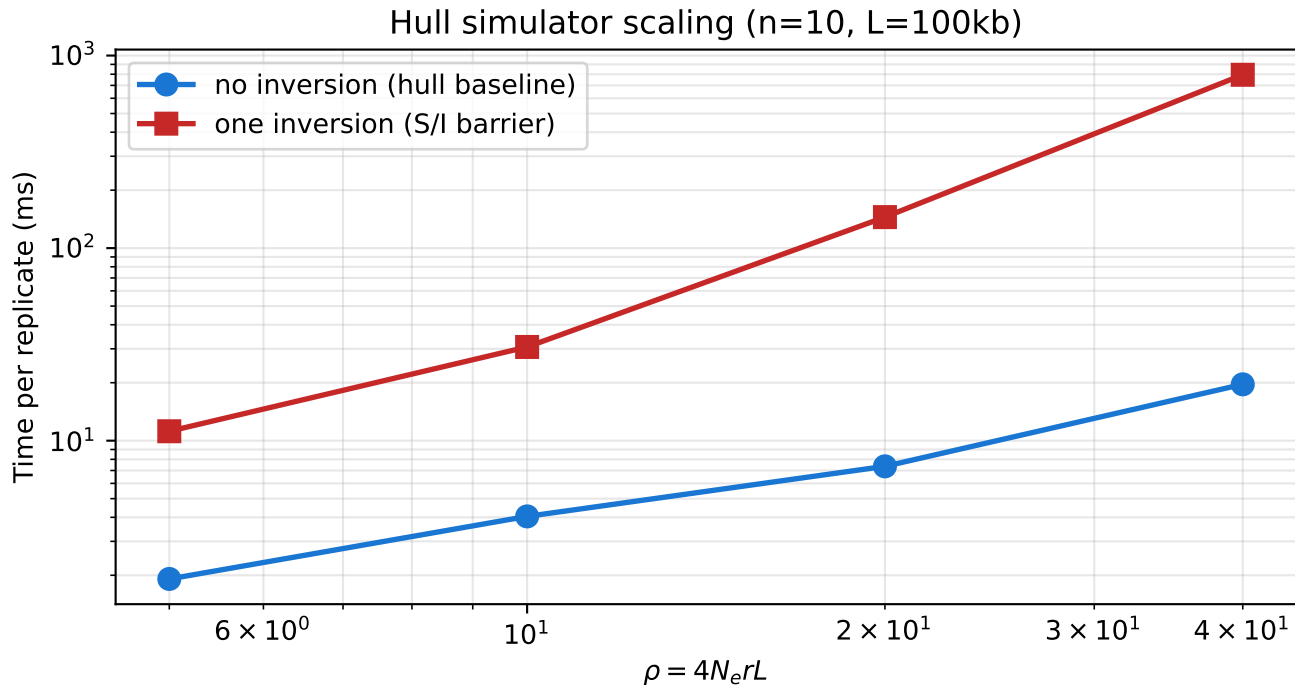


Figure 7. Hull simulator performance scaling with recombination rate. Wall-clock time per replicate (ms) for 10 haploid samples on a 100 kb sequence. Blue: no inversion (baseline panmictic coalescent). Red: one inversion with S/I karyotype barrier (bp_left=30kb, bp_right=70kb, t_inv=100,000 gen). The inversion adds overhead from per-segment class tracking and gene-flux events, but cost scales sublinearly with ρ .

Parameters: $N_e=10,000$, $L=100$ kb, $n=10$, 20 replicates per point.

Command: `pixi run -e all python examples/make_figures.py`